Course Overview and Starting Data Analytics

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WEEK 1 LECTURE DECK

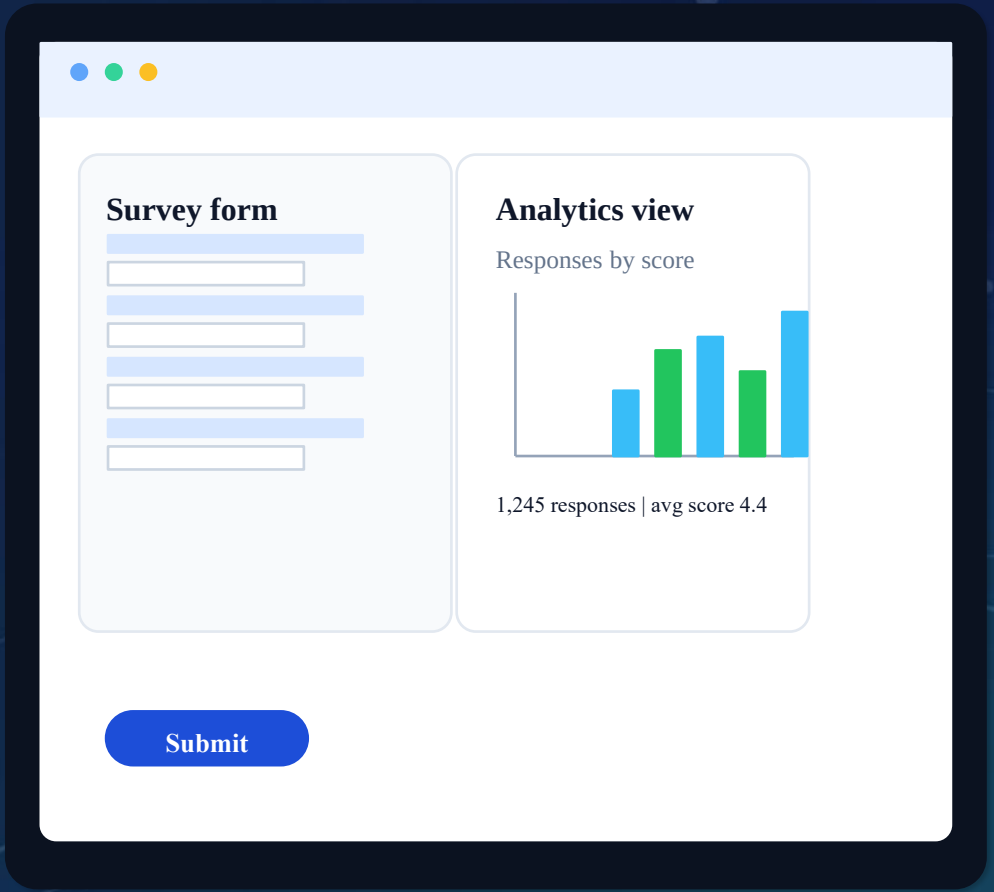
Course Overview and Starting Data Analytics

Topics for this week:

- data analytics lifecycle
- tools setup: Python, pandas, Flask, MySQL
- CSV dataset introduction
- Python refresh and pandas basics

Sample project: Online Survey System

Use simple code, real data, and a small dashboard to learn step by step.



From survey form to dashboard

collect → store → clean → analyze → show

Week 1 roadmap

What students should understand and do by the end of this week

1. Course overview
2. Lifecycle
3. Tools setup
4. CSV dataset
5. Python refresh
6. pandas basics

Knowledge

Students can explain what data analytics is and name the main steps in the lifecycle.

Skills

Students can install tools, open a CSV file, and run basic Python and pandas commands.

Mindset

Start with a real question, not with random code.

What this course is really about

Simple idea: use data to answer a useful question

Input

- Survey responses
- CSV files
- Database tables
- Time, rating, device, location

Work

- Clean missing data
- Group and count results
- Find patterns
- Build simple charts and pages

Value

- Know user opinion
- Find weak points
- Track KPIs
- Support better decisions

Example question for our sample project: "Which province, age group, or device has the lowest satisfaction score in the online survey?"

Question first

Semester sample project: Online Survey System

A small but complete system that connects data collection and analytics

1. User answers survey	Questions about service, satisfaction, rating, device, province
2. Flask receives data	The web app accepts form input and checks basic rules
3. MySQL stores records	Each response is saved as a row in a database table
4. pandas reads for analytics	We load data to create KPI tables, charts, and insights

Why this project is good for this course

- It has real user data, not only toy examples.
- It lets students practice CSV, SQL, and web output together.
- It is small enough for a semester, but complete enough to feel real.
- It creates a clear story from data collection to dashboard insight.

Real project **Full flow** **Good for teamwork**

SESSION

Session 1: Course idea, lifecycle, and tools

We begin with the big picture before we write code.

The data analytics lifecycle

A simple step-by-step path we will use again and again

1	2	3	4	5	6
Ask	**Collect**	**Clean**	**Analyze**	**Visualize**	**Decide**
Define the business question. Example: Why are users unhappy?	Get survey answers from forms, CSV files, or databases.	Fix missing values, wrong types, and duplicate rows.	Count, compare, group, and find patterns.	Turn numbers into tables and charts people can read.	Use the result to improve service or product action.

Survey example: “Users from Province A report lower satisfaction on mobile devices. We now investigate the mobile experience.”

Tool stack for this course

Each tool has one clear job in our online survey project

Flask dashboard Show pages, tables, and charts in a web app

pandas analysis Read data, clean it, group it, and compute KPIs

MySQL storage Save survey responses in structured tables

CSV files Simple file format for import, export, and quick practice

Python language The base language that connects all parts together

Simple pipeline

Survey form → MySQL table → pandas notebook/script → Flask page

- You do not need to master everything in one day.
- This week, only learn the purpose of each tool.
- Later weeks will go deeper into SQL, Flask, and project integration.

Class setup: first working environment

Keep it simple: one folder, one terminal, one short test

Suggested setup steps

- Install Python 3 on your computer.
- Create a course folder, for example: data_analytics_course
- Open VS Code or another code editor.
- Open Terminal / Command Prompt in that folder.
- **Install libraries:** pandas, Flask, SQLAlchemy, mysql-connector-python
- Run a short Python test to check everything works.

Example commands

```
python --version
pip install pandas Flask SQLAlchemy mysql-connector-python
python
>>> import pandas as pd
>>> print("Ready")
```

Target result: the system prints “Ready” with no error.

Meet the starter dataset: survey_responses.csv

A CSV file is a simple text file where each row is one response

Sample rows

response_id	submit_time	province	age_group	score	device	recommend
1001	2026-01-12 09:10	Phnom Penh	18-24	5	mobile	yes
1002	2026-01-12 09:15	Kandal	25-34	4	desktop	yes
1003	2026-01-12 09:18	Siem Reap	18-24	2	mobile	no
1004	2026-01-12 09:28	Battambang	35-44	3	tablet	no

How to read it

- Each row = one survey response
- Each column = one variable / field
- The first row usually holds column names
- CSV is easy to open in Excel, Python, or many other tools
- Bad data can still exist, so we must check quality later

SESSION

Session 2: Python refresh and pandas basics

Now we move from concepts to first commands.

Python refresh for analytics

We only need a few core ideas to begin

Small Python example

```
scores = [5, 4, 2, 3]
user = {"province": "Phnom Penh", "device": "mobile"}

for s in scores:
    if s < 3:
        print("Low score:", s)

print(user["device"])
```

What to remember

- **List:** store many values in order
- **Dictionary:** store key → value pairs
- **for loop:** repeat work for many items
- **if statement:** make a simple decision
- **print():** quick way to see a result while learning

Enough for today

Learn by doing

pandas basics: what is a DataFrame?

A DataFrame is like a smart table inside Python

Think of a DataFrame as:

- rows = records or observations
- columns = variables or fields
- index = row labels
- one object that holds the whole table
- easy to filter, group, count, and summarize

Mental picture

index	province	score	device
0	Phnom Penh	5	mobile
1	Kandal	4	desktop
2	Siem Reap	2	mobile
3	Battambang	3	tablet

This table lives inside Python, so we can work with it using code.

First pandas commands to know

read_csv(), head(), info(), and describe()

Code

```
import pandas as pd

df = pd.read_csv("survey_responses.csv")
print(df.head())
print(df.info())
print(df.describe())
```

What each command does

- read_csv(): load CSV into a DataFrame
- head(): show the first few rows
- info(): show column names, data types, and non-null counts
- describe(): show summary numbers for numeric columns

Tip: Students should predict the output before they run the code.

Week 1 activities, exit task, and recap

End the week with one small success

In-class activity

- Open the sample CSV file
- Run read_csv()
- Show head() and info()
- Write one sentence about what the dataset contains

Exit task

- Explain the lifecycle in your own words
- Name the job of Python, pandas, Flask, and MySQL
- Tell the class what one row and one column mean in the survey CSV

By the end of Week 1

- Students know the course direction
- Students can explain the lifecycle
- Students can load a CSV file in pandas

Remember: small, clear steps are better than rushing.

Thanks for attention,

Any Question?